Boundary-Aware Static Robot Deployment: Sampled Safety and Limits of Transport Preparation

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***Abstract*—Placing robots around an object is a distinct problem from reliably transporting it. We study boundary-aware deployment through limited-range centroidal targets and a velocity safety filter, separating an exact static oracle-map analysis from locally sensed implementations and downstream contact experiments. For a fixed polygon and single-integrator robots, a feature-cover construction gives whole-hold safety under explicit initialization, communication, sampling and projection assumptions. A projected-residual bound clarifies why low commanded motion need not imply complete coverage. In a frozen 169-episode campaign, a local segment candidate passes nine of nine static path checks, versus one of nine for the original point-sample implementation, while reducing the measured deployment cost. Its gap guard intervenes on 5,315 of 5,400 frames. Transport completion remains limited: boundary CVT completes 5/12 new-seed tasks and 1/18 perturbed tasks. An independent contact-engine case also fails. These results support a restricted static deployment claim and motivate, rather than establish, reliable transport with unknown moving boundaries.**

***Index Terms*—Multi-robot deployment, coverage control, sampled safety, cooperative transport.**

I. Introduction

Robots that must push an unfamiliar object first need useful positions around its boundary. A team can approach the boundary quickly while leaving important regions unsupported; a configuration can also appear stationary because safety constraints block its nominal motion. Once contact starts, estimation drift, asymmetric forces and release timing introduce further failure modes. A single success label obscures these distinctions.

We investigate a boundary-aware deployment controller with separate information permissions for control and evaluation. Limited-range Voronoi cells allocate observed boundary mass, while a safety filter modifies the resulting velocity. The theoretical reference uses a fixed known polygon. The local implementation receives only sensed or relayed boundary information; true geometry is reserved for an external evaluator. Contact transport is evaluated as a downstream task, without extending the static theorem to moving objects.

The contributions are an explicit static feature-cover safety argument, a residual interpretation for the saturated-then-projected centroidal law, and an auditable comparison that retains failed and censored episodes. The empirical segment repair improves the observed static paths with essentially unchanged cost reduction, but its conservatism and incomplete geometry prevent a local-map safety theorem. Preparation and transport results are reported separately. We do not claim general transport superiority or engine invariance.

II. Related Work

Voronoi-based coverage control connects locational costs to distributed centroidal motion [1]. We use that connection to allocate boundary density, with truncated cells and a fixed-density theoretical reference. Safety projection and numerical centroid errors change the interpretation of the classical centroidal equilibrium; coverage convergence cannot simply be inherited.

Control barrier functions express forward-invariance requirements as constraints on admissible controls and can be combined with performance objectives in a quadratic program [2]. Our concern is the held command between samples, including switches of the closest polygon feature. A feasible instantaneous inequality against an incomplete observed boundary is insufficient.

Caging-based decentralized manipulation maintains object closure while moving toward a goal [3]. Sequential caging has also addressed unknown object shapes, distributing contact locations before transport and evaluating both simulation and physical robots [4]. Those results make unknown-shape transport an established research direction, not a novelty claim of the present work. Here the narrower question is what a boundary-density deployment objective and its static safety filter actually establish before dynamic contact. We do not equate proximity to a boundary with object closure or sufficient contact wrench authority.

III. Model, Information and Objectives

Let $D \subset \mathbb{R}^2$ be a closed rectangle and $S \subset D$ a fixed simple polygon. There are $N = 16$ robots with $\dot{p}_i = u_i$, $\|u_i\| \le u_{\max}$. At $t_k = k\Delta$, all controls are computed from the same frozen configuration P_k, then held for $\tau \in [0, \Delta]$. The theoretical oracle exposes the polygon features. Local arms instead construct their maps from range observations and one-hop communication. An independent evaluator may access S and simulator object states, but those values do not select local control commands.

For the analysis only, fix a nonnegative reference density $\phi^\star$ around a uniform boundary offset. Direction-dependent, exploration and changing-map terms are excluded. With $\Omega_i = V_i(P) \cap B(p_i, R)$, define

$$H^\star(P) = \int_D \min\{R^2, \min_i \|q - p_i\|^2\}\phi^\star(q)\, dq. \qquad (1)$$

For each positive-mass cell, $m_i^\star = \int_{\Omega_i} \phi^\star$ and $c_i^\star = (m_i^\star)^{-1} \int_{\Omega_i} q\phi^\star(q)\, dq$. The implementation computes $\hat{m}_i, \hat{c}_i$,

projects an out-of-disk centroid onto the local disk and uses p_i when its numerical mass is below the floor. The command is

$$v_i = \mathrm{sat}_{u_{\max}}(k_c(\hat{c}_i - p_i)), \qquad U_i = \Pi_{F_{i,k}}(v_i). \tag{2}$$

Here $F_{i,k}$ is the intersection of the speed ball, shared-responsibility robot rows, rectangular wall rows and object-feature rows. The exact analysis uses the original robust margin, with no infeasibility scaling or clamping.

Static quality is measured by cost reduction and displacement as well as safety. In transport, controller CONTACT_READY, a dwell of positive forward physical contacts, G500, settling and final task completion are distinct events. Completion requires the existing settling rule and final Euclidean position error at most the task tolerance, 0.12 m in the reported transport cases. An along-track arrival estimate alone does not satisfy this definition.

IV. Static Sampled Safety and Residual Interpretation

For a polygon feature F (a closed edge segment), set $h_F(p) = \mathrm{dist}(p, F) - r_{\mathrm{safe}}$. Include every feature in

$$\begin{aligned}\Phi(p) = \{F :\ & \mathrm{dist}(p, F) \le d_\partial(p) + 2u_{\max}\Delta, \\ & \mathrm{dist}(p, F) \le R_{\mathrm{row}}\},\end{aligned} \tag{3}$$

where d_∂ is distance to the polygon boundary. Impose

$$\nabla h_F(p)^T u \ge -\gamma_{\mathrm{obj}} h_F(p) + \rho. \tag{4}$$

Distances are to finite segments, not their infinite supporting lines.

A. Assumptions and whole-hold argument

The initial configuration lies in D, has pair separation at least $d_{\min}$, and lies outside S with signed clearance at least $r_{\mathrm{safe}} + \rho/\gamma_{\mathrm{obj}}$. Geometry and density are fixed; communication has no dropout and range at least $2R$, including all pairs within $d_{\min} + 2u_{\max}\Delta$. The robot rows use the same frozen positions and symmetric responsibility. Require $\gamma_{\mathrm{obj}}\Delta \le 1$, $\gamma_{\mathrm{agent}}\Delta \le 1$, and

$$R_{\mathrm{row}} - u_{\max}\Delta \ge r_{\mathrm{safe}} + \rho/\gamma_{\mathrm{obj}}. \tag{5}$$

All projections and geometry in the following argument use exact arithmetic.

Proposition 1 (Static safety under the stated assumptions): The exact cascade preserves object clearance $r_{\mathrm{safe}} + \rho/\gamma_{\mathrm{obj}}$, pair separation $d_{\min}$ and membership in D throughout every hold. Its feasible velocity sets contain zero at every sample, so their Euclidean projections exist recursively.

Proof: Distance to a closed convex segment is convex. For each included feature, (4) and the supporting-gradient inequality give

$$h_F(p + \tau u) \ge (1 - \gamma_{\mathrm{obj}}\tau) h_F(p) + \rho\tau. \tag{6}$$

An in-range feature excluded by the cover is initially more than $2u_{\max}\Delta$ beyond the nearest boundary distance. Its distance decreases by at most $u_{\max}\Delta$, by the Lipschitz property. Range-excluded features remain beyond the required margin by (5). Thus the minimum boundary distance retains the margin even at feature switches or when the included set is empty. Starting outside, continuity excludes a first boundary crossing. For included pairs, the sum of the two responsibility inequalities bounds the initial relative-distance derivative; convexity of squared distance and $\gamma_{\mathrm{agent}}\Delta \le 1$ preserve separation. Omitted pairs cannot close the initial excess distance within a hold. Rectangular wall constraints preserve each coordinate throughout the segment. These margins make zero feasible in every stacked row at the next sample. Induction proves the assertion. ∎

The reference parameters are $\Delta = 0.05$ s, $k_c = 0.9$, $u_{\max} = 0.35$ m/s, $R = 0.8$ m, $R_{\mathrm{row}} = 0.60$ m, $d_{\min} = 0.28$ m, $r_{\mathrm{safe}} = 0.065$ m, $\rho = 0.02$, $\gamma_{\mathrm{obj}} = 8\ \mathrm{s}^{-1}$ and $\gamma_{\mathrm{agent}} = 6\ \mathrm{s}^{-1}$. The object margin is 67.5 mm; the local empirical path acceptance threshold is 65 mm. These are different quantities. Full derivations and parameter ledgers accompany the source.

B. Cost dissipation does not imply complete coverage

Define $J_k = \sum_i m^\star_{i,k} \|U_{i,k}\|^2$ and $E_k = \sum_i m^\star_{i,k} \left\|\hat{c}_{i,k} - c^\star_{i,k}\right\|^2$, with zero-mass cells contributing zero. Freezing the current partition gives a quadratic majorant of $H^\star$; reassigning points to their nearest truncated cell only reduces that majorant. Projection optimality with zero feasible then yields, for $a = 2/k_c - \Delta > 0$,

$$\overline{J} \le \frac{2H^\star(P_0)}{aK\Delta} + \frac{4\overline{E}}{a^2}. \tag{7}$$

To see this, write $v_i = \lambda_i(\hat{c}_i - p_i)$, with $0 < \lambda_i \le k_c$. Projection optimality and zero feasibility imply $v_i^T U_i \ge \|U_i\|^2$. The partition majorant then gives $H^\star(P_{k+1}) - H^\star(P_k) \le -\Delta a J_k + 2\Delta\sqrt{E_k J_k}$. Young's inequality bounds the right-hand side by $-\Delta a J_k/2 + 2\Delta E_k/a$; summing and using $H^\star \ge 0$ proves (7). This argument does not assume numerical monotonicity of the observed cost.

To clarify its meaning, let r_i be (2) computed with $c^\star_i$ in place of $\hat{c}_i$ and the same $F_{i,k}$. Nonexpansiveness of saturation and projection gives $\|r_i - U_i\| \le k_c \|c^\star_i - \hat{c}_i\|$. Hence

$$G_k := \sum_i m^\star_{i,k} \|r_i\|^2 \le 2J_k + 2k_c^2 E_k, \tag{8}$$

and (7) bounds the average exact-centroid projected residual. Zero residual means no descent through the filter's frozen feasible velocity set. It does not imply stationarity of the original geometric-position problem without an additional tangent-cone equivalence. Nearly massless cells can also hide poor spatial support. Therefore neither a small J nor a small G establishes full boundary coverage, useful contact capacity or transport success.

C. Local maps, arithmetic and moving boundaries

A sufficient local-map extension would require certified coverage of every true boundary point relevant to the hold tube within error ϵ, with omitted features independently excluded. A whole-hold estimated clearance of at least $r_{\mathrm{safe}} + \epsilon$ would then transfer to true clearance by the triangle inequality. Our

TABLE I
FROZEN V2 STATIC PATHS: NINE CASES PER ARM; REQUIRED CLEARANCE 65 MM.

Arm	Safe paths	Min. (mm)	Mean H_f/H_0
Original local samples	1/9	48.33	0.1533
Segment candidate	9/9	66.04	0.1544
Oracle reference	9/9	67.50	0.1534

TABLE II
FROZEN V2 TASK COMPLETION, INCLUDING FAILED AND CENSORED TASKS.

Population	CVT	Nearest	Spacing
Development	5/9	5/9	4/9
New-seed generalization	5/12	3/12	1/12
Perturbation robustness	1/18	2/18	1/18

sensor does not provide this coverage certificate. In particular, clipping a measured observation gap does not certify missing geometry.

Floating-point row residuals also require a separate budget. If the physical inequality has violation at most ϵ_u, the last term in (6) becomes $(\rho-\epsilon_u)\tau$. Row coefficient errors, state error and tracking error contribute in addition to solver tolerance. Thus a floating-point `optimal` status is not an exact-arithmetic certificate. Moving polygons would further require swept-feature bounds, uncertain boundary velocities and recursive feasibility under contact dynamics. Those extensions remain open.

V. EXPERIMENTS AND FAILURE EVIDENCE

A. Protocol and provenance

The frozen draft-v2 campaign contains 169 planned episodes: 27 static comparisons, 27 development transport comparisons, 15 target ablations, six active-exploration diagnostics, 36 generalization tasks, 54 perturbation tasks, two QP references and two contact-engine references. All planned cases remain in their denominators. The source snapshot, expanded configurations, initial-state identities and product hashes are archived. The implementation audit passes with zero execution errors; that gate does not require scientific task success. Two serial repeats, excluded from scientific denominators, reproduce their parallel results.

Static comparisons use rectangle, L and C polygons and seeds 2, 5 and 8, with 600 complete control holds each. The oracle, original local samples and local segment candidate share the task and initial state. Transport compares boundary CVT, nearest observed targets and local spacing; only the pre-CONTACT_READY deployment target differs. Generalization uses 12 new seeds with the frozen layout/scale factors. Perturbations use two independent seeds repeated across shapes and noise, communication loss and slower sensing. These are two seed clusters, not 18 independent replications per method.

B. Static deployment

Table I shows that the local segment candidate passes all nine whole-hold path checks; the original local point formulation violates eight. The segment candidate's smallest observed margin over the threshold is only 1.04 mm. Mean robot displacement per episode ranges from 0.810 to 1.013 m, so the cost reduction does not result from remaining stationary. However, the gap guard truncates 5,315 of 5,400 frames. This frequent intervention and the limited three-shape geometry coverage prevent broad generalization.

The independent planar-QP and CVXPY references both pass their 600-frame checks, with maximum recorded position difference 1.82 mm. We report agreement of the checked safety outcomes, not equality of numerical trajectories. Finite path verification remains distinct from the exact proposition's assumptions.

C. Preparation versus final transport

In generalization, CVT reaches controller readiness 1.364 s earlier on average than nearest-target deployment over 11 complete pairs; one pair is censored. The exploratory seed-cluster bootstrap interval for CVT minus nearest is $[-2.736, -0.145]$ s. Against local spacing, the difference is -1.632 s with interval $[-6.668, 1.582]$ s. Complete-pair selection can bias readiness differences. Neither comparison establishes transport reliability; the actual task rates are given in Table II. Development and robustness have only three and two independent seed clusters, respectively, so their rates are descriptive and receive no inferential interval.

The main exploration gain is zero, and removing it produces identical replay arrays in all three baseline shapes. It is a null intervention, not evidence for the contribution of exploration. A separate gain-six diagnostic has mixed outcomes. One seed per shape is insufficient for an ablation benefit claim. In the independent-engine pair, penalty contact completes at 0.111 m endpoint error; Pymunk reaches 0.378 m and triggers its watchdog. Successful recording and execution in both engines do not establish physical-model invariance.

D. Failure classification and reliability repairs

Reanalysis of the 140 transport receipts retains the frozen outcomes and uses each task's original tolerance. For CVT development failures, three are dominated by along-track undershoot and one by overshoot. For its 17 robustness failures, eight exceed tolerance in the final transverse component, eight exceed it in both transverse and longitudinal components, and one is longitudinal undershoot. These are geometric outcome classes. Missing physical-contact dwell and missing phase events are recorded separately and are not assigned as causal mechanisms from compact receipts alone.

Development inspection also found that slow planning could retain a nominal command across a supervisor change, and a robot whose progress estimate lagged the team could still enter the push set after HOLD. The default revision invalidates held commands when supervisor permissions change and explicitly prohibits pushing in HOLD. More aggressive changes to brake-integral release and feedback routing were tested but

caused development regressions; they remain experimental options disabled by default. This negative result cautions against equating a plausible control-law change with improved task reliability. No revision changes tolerances, horizons or the frozen results above. Regression tests establish specific control responses; development replays are separate from independent validation and do not establish robust transport.

VI. Limitations and Conclusion

The strongest conclusion is static sampled deployment under explicit fixed-map assumptions, supported by independent path observations for a local segment candidate on nine cases. Static deployment quality and downstream controller readiness provide useful measurements, but neither establishes caging or contact wrench sufficiency. Robustness failures and the Pymunk discrepancy are material limits, not execution errors to exclude.

Further validation must freeze additional geometry, occlusion and sampling conditions before observing outcomes, report gap-guard intervention alongside deployment quality, and use independent seed clusters for transport comparisons. Mechanism studies need frame traces for estimate bias, contact-loss response, brake entry and terminal error. Dynamic safety and hardware deployment require additional modeling and sensing evidence. The present results delimit a static deployment contribution while identifying the measurements needed to assess reliable transport in subsequent work.

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